$$\frac{\mathrm{d}}{\mathrm{d}t}\langle\hat{P}\rangle_t = \frac{\mathrm{d}}{\mathrm{d}t}\mathrm{Tr}_{\mathcal{S}}\left\{\hat{\rho}_{\mathcal{S}}(t)\hat{P}\right\} = \mathrm{Tr}_{\mathcal{S}}\left\{\frac{\mathrm{d}\hat{\rho}_{\mathcal{S}}(t)}{\mathrm{d}t}\hat{P}\right\}$$
$$= -M(\Omega^2 + \widetilde{\Omega}^2)\langle\hat{X}\rangle_t - 2\gamma\langle\hat{P}\rangle_t. \tag{5.32}$$